

STATEMENT OF PURPOSE

Minjun Kim | Graduate Application in Electrical Engineering and Computer Science

The first time I watched a scholarship applicant abandon a form because she could not determine whether a clause applied to her, I saw a computer science problem hidden inside an administrative one. The documents contained the answer, but their structure, language, and exceptions made that answer inaccessible. Since then, my goal has been to build machine learning systems that can read complex documents, communicate uncertainty, and help people act with confidence.

At the Hanul Institute of Technology, I pursued this goal through a Computer Engineering curriculum spanning algorithms, systems, probability, and machine learning. Coursework taught me to move between abstraction and implementation: I analyzed complexity in Algorithms, built a scheduler in Operating Systems, and trained multimodal models in Computer Vision. My 3.92/4.00 GPA reflects consistent preparation, but the most important outcome was learning to test an idea against both quantitative evidence and real user behavior.

In the Intelligent Systems Laboratory, advised by Professor Jisoo Lee, I developed a layout-aware document understanding pipeline. Early experiments achieved strong extraction accuracy on clean forms but failed on multi-column notices and scanned appendices. I created an error taxonomy, added visual-position features, and introduced a calibrated confidence layer. On our 12,000-page benchmark, the resulting system raised field extraction F1 from 0.84 to 0.91. More importantly, it exposed low-confidence evidence spans to reviewers instead of presenting uncertain predictions as facts. This work led to a workshop paper and shaped my interest in trustworthy, human-centered AI.

I extended this perspective during a summer internship at Mirae AI Research. There I built retrieval and reranking components for bilingual technical search. By separating retrieval failures from generation failures and introducing hard-negative mining, our team reduced unsupported answers by 19% in offline evaluation. Working with researchers and product engineers taught me that reliable AI depends not only on model quality, but also on measurement design, data provenance, and interfaces that help users recognize limitations.

STATEMENT OF PURPOSE

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My capstone project, ApplyWise, brought these lessons together. Our four-person team built a macOS document agent that compares application requirements with personal evidence, classifies eligibility, highlights exact source passages, and organizes submission files. I led the model and interaction design, while also implementing the evidence graph and evaluation harness. Pilot users completed a representative application checklist 34% faster and made fewer requirement-matching errors. The project convinced me that document intelligence can have meaningful social impact when prediction, explanation, and workflow design are treated as one system.

For graduate study, I want to investigate three connected questions: how multimodal models can reason over long, heterogeneous documents; how uncertainty can be calibrated and communicated at the evidence level; and how smaller, efficient models can support private on-device workflows. MIT's culture of integrating machine learning, systems, and human-computer interaction is especially compelling for this agenda. I am drawn to an environment where rigorous model research can be tested through real tools and evaluated with the people who depend on them.

I would contribute experience in document AI, bilingual retrieval, Apple-platform development, and reproducible experimentation. I also bring a collaborative approach shaped by leading Hanul's AI student society and teaching data structures. In graduate school, I hope to work across research groups, publish careful empirical studies, and build open evaluation resources for trustworthy document systems.

My long-term aim is to lead research that makes high-stakes information understandable and actionable without concealing uncertainty. Graduate training at MIT would give me the theoretical depth, interdisciplinary community, and research discipline to pursue that aim. I am prepared to contribute as an engineer, researcher, and colleague, and I look forward to developing systems that earn trust through evidence rather than appearance.